Natural Language Processing

Introduction

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Session 01



Take Home

- What is natural language processing about?
- What do we cover in this class?
- What are the formalities?

Outline

1 Motivation

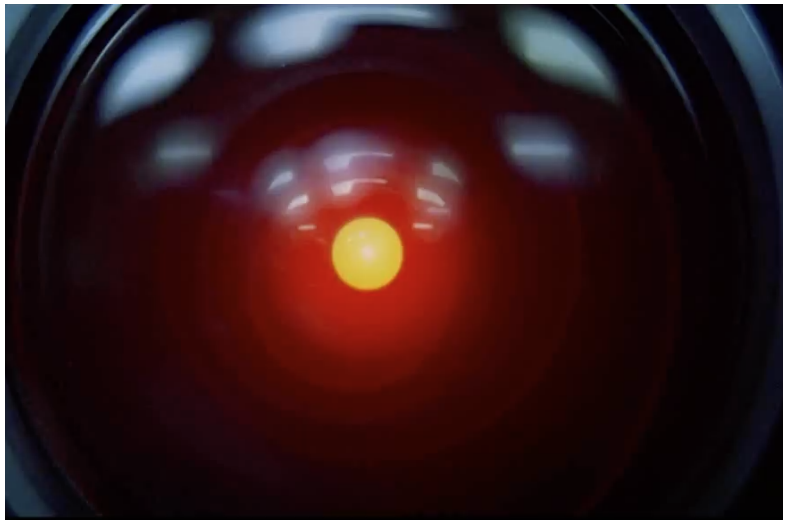
2 Formalities

3 Topics

HAL in Odyssey 2001

- Which skills does the computer have in the following excerpt from the 1968 movie "2001: A Space Odyssey"?
- Which of these skills do computers have nowadays? Where do they lack?
- (after we watched the video, we'll collect our observations)

`https://www.youtube.com/watch?v=ARJ8cAGm6JE`



Skills of HAL

Talk
Able to understand the person
Phonetics, phonology
Grammar
Word meaning, semantics
Utilitarian thinking – intentions
Reject orders
Lip reading
Grounding entities mentioned in language in the world
Non-literal meaning, irony

What is Natural Language Understanding?

We try to make computers 'understand' natural language



Understanding an utterance means to:

- determine its truth
- react to it in a suitable way
- determine its entailments
- translate it into another language
- ...

What is Natural Language Understanding? (Cont'd)

Often, the term is used to contrast with
natural language generation

Natural Language Understanding

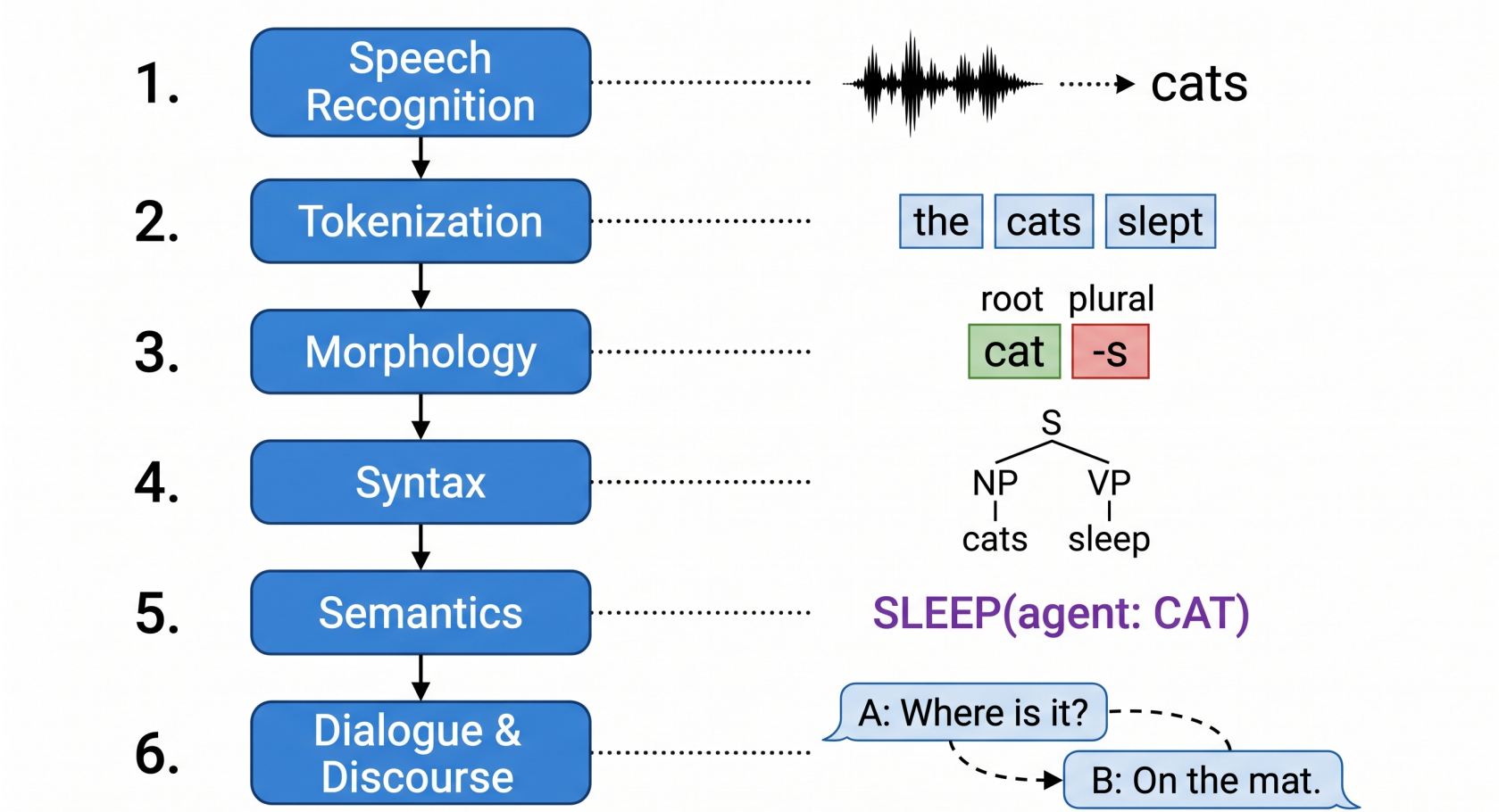
$$\text{Text} \Longrightarrow \text{Analysis}$$

Natural Language Generation

$$\text{Text or Data} \Longrightarrow \text{Text}$$

(we do not focus on generation in this course)

Natural Language Processing



Rough Overview of Progress in the Field



- 1960s: Pattern-matching using hand-built rules
- 1970-80s: Linguistically rich, logically-informed systems, usually grounded in highly structured databases
- 1990s: "statistical revolution" in NLP: decrease in NLU work
- 2010s: NLU uses a mix of techniques from previous decades
- 2015: Deep learning claims feature engineering is not required
- 2020: Prompting with LLM combines deep learning, rules and feature engineering

1. Pattern-Matching Using Hand-Built Rules

ELIZA (Weizenbaum, 1966)



- ELIZA the Rogerian Psychotherapist
  - Person-centered therapy, after Carl Rogers
- A very simple program written by Joseph Weizenbaum in 1966
- ELIZA does not need to know anything about reality: hand-built rules

How does ELIZA Work?

Some rules for ELIZA:

- 'my' $\rightarrow$ 'your' and 'I'm' $\rightarrow$ 'you are'
- 'I am (depressed|sad)'
  $\rightarrow$ 'I am sorry to hear you are (depressed|sad)'
- If the utterance contains 'always'
  $\rightarrow$ 'Can you think of a specific example?'
- If the utterance contains 'all' $\rightarrow$ 'In what way?'

ELIZA: Example

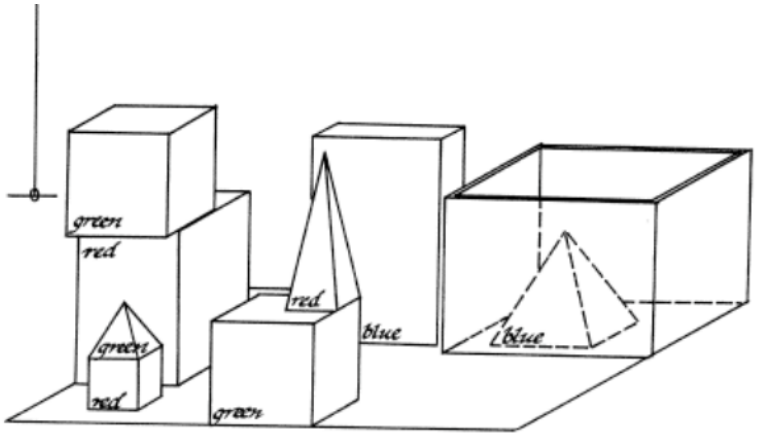
- **User:** Men are all alike
- **ELIZA:** In what way?
- **User:** They're always bugging us about something or other.
- **ELIZA:** Can you think of a specific example?
- **User:** Well, my boyfriend made me come here.
- **ELIZA:** Your boyfriend made you come here.
- **User:** He says I'm depressed much of the time.
- **ELIZA:** I am sorry to hear you are depressed.

(run Eliza, e.g., in emacs with `M-x doctor`)

2. Linguistically Rich, but a Small World

SHRDLU (Winograd, 1971)

The user interacts with the system in order to move various objects around in a "blocks world"



- The world is small: only 50 words are needed to describe it
- It keeps track of the user's previous moves
- The user can give names to things
- The program could answer questions about what is possible in the world

Example from SHRDLU

[run video]

Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BOX.
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?
Computer: YES.

Example from SHRDLU

[run video]

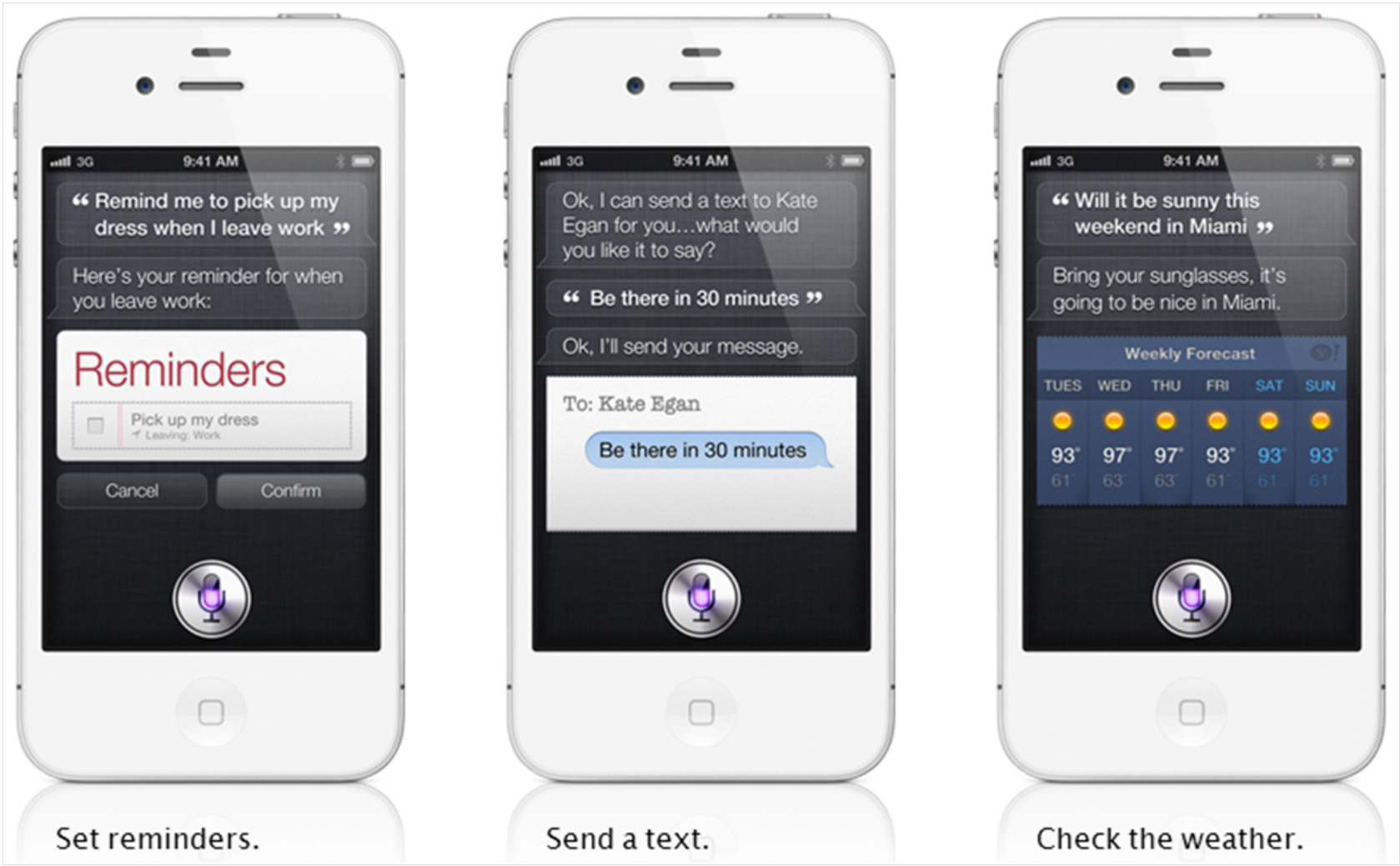
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Computer: YES.
...

3./4. The Statistical Revolution and the New Renaissance

- The statistical revolution in NLP in the mid 90s
- This meant a little bit less attention for NLU: they had to start again from the beginning!
- As a result: a mix of linguistically-rich, logical and statistical approaches
- Now: Deep learning and large language models, which is sometimes considered the next revolution.

Natural language processing until roughly last year

e.g. Apple Siri



The Promise of Siri



Where is The Hobbit playing in Mountain View?

The Hobbit is playing at the Century 16 Theater.

When is it playing there?

It's playing at 2pm, 5pm, and 8pm.

OK. I'd like 1 adult and 2 children for the first show. How much would that cost?



Need domain knowledge, discourse knowledge, world knowledge

The Promise of Siri



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Need domain knowledge, discourse knowledge, world knowledge (ChatGPT, Gemini etc., are partially fulfilling this promise now, Google assistant and Siri are just starting.)

The Reality of Siri (when Siri came out)



- **Colbert:** Write the show.
- **Siri:** What would you like to search for?
- **Colbert:** I don't want to search for anything! I want to write the show!
- **Siri:** Searching the Web for "search for anything. I want to write the shuffle."
- [...]
- **Colbert:** For the love of God, the cameras are on, give me something?
- **Siri:** What kind of place are you looking for? Camera stores or churches?

NLP Today: a Growing Interest

In different fields of academia and industry:

- voice-controlled assistants (Siri, Google Now, Cortana)
- natural language search (Google, Bing)
- question answering
  (IBM's Watson, Wolfram Alpha, ChatGPT, Alpaca)
- social media analytics
- computational literary studies
- ...

The Goal of This Course

- We want to give you an overview of challenges in NLP
- We provide example methods to solve these challenges
- We do not reach state of the art methods here, but we provide the fundament for them.
- Complemented by other courses:
  IRTM, DL4NLP, PGM4NLP, ILT, EA, Seminars, Projects

Outline

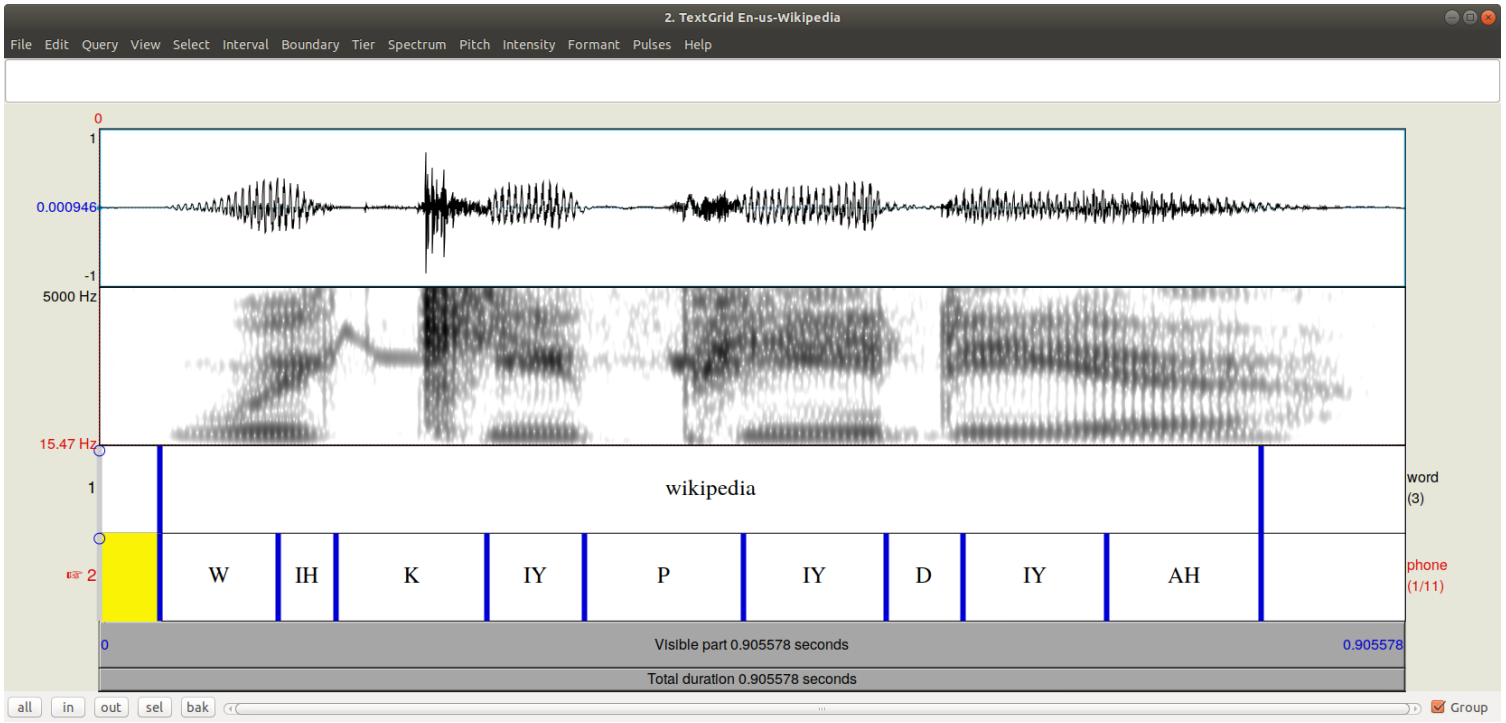
1 Motivation

2 Formalities

3 Topics

Phonetics

Reconstruct a sequence of words from an audio signal



Preprocessing and Tokenization

Go from text in the wild to text that machines can process



THE TIMES

Is your home toxic? The scientists' view

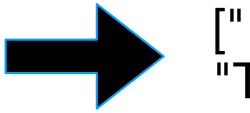
'We earn £230k but our childcare is a tight squeeze'

King will not fund Andrew's case against police over his arrest

PM looks to push back decision on defence cash

But Tories to pledge £10bn for missile shield

Novelist and former MP Jeffrey Archer dies aged 86



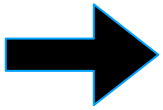
["Is", "your", "home", "toxic", "?",
"The", "scientist", "'s", "view"]

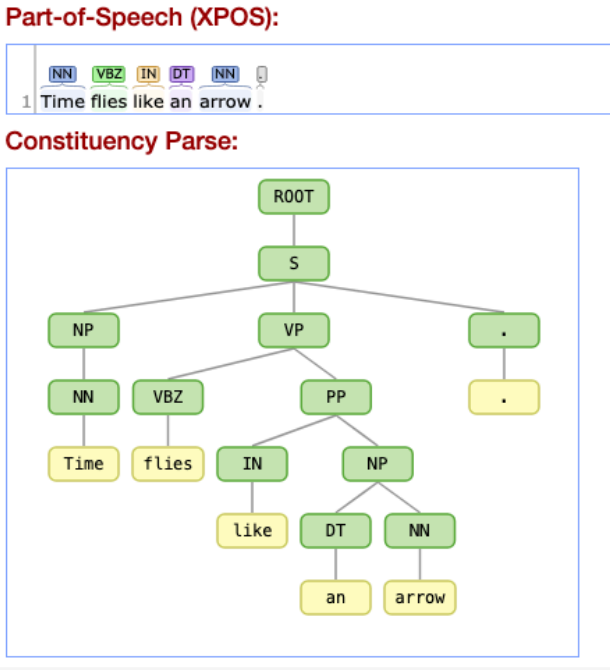
Part of Speech Tagging and Constituency Parsing

From sentences to structured representations

Time flies like an arrow.

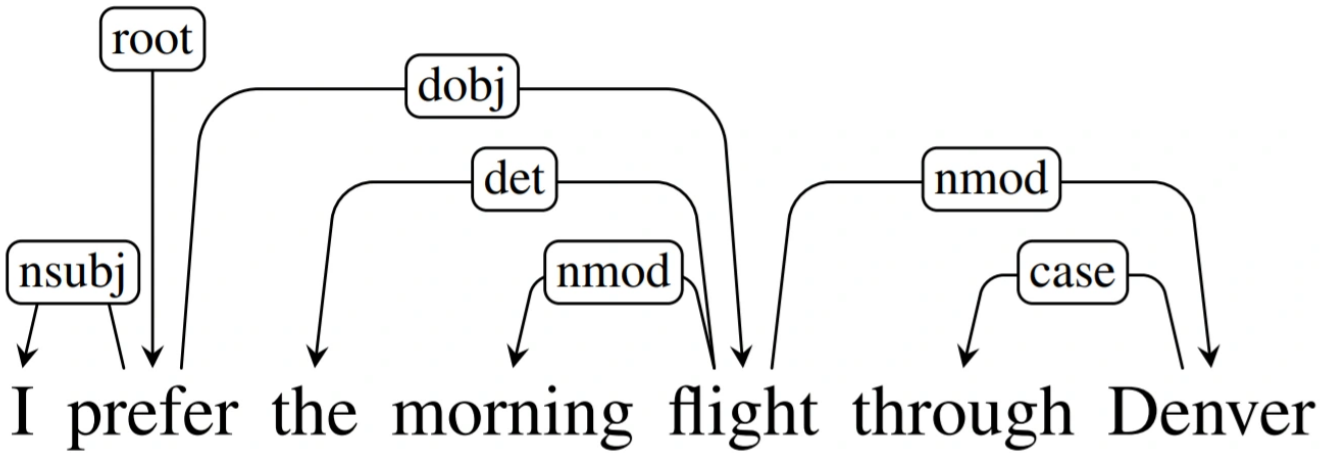
Fruit flies like a banana.



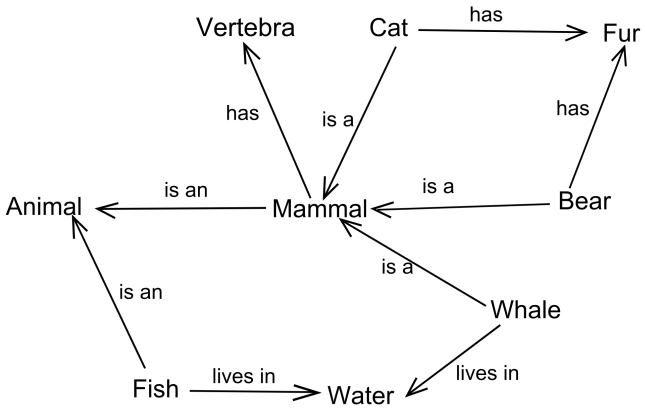


Dependency Parsing

. . . and more structured representations

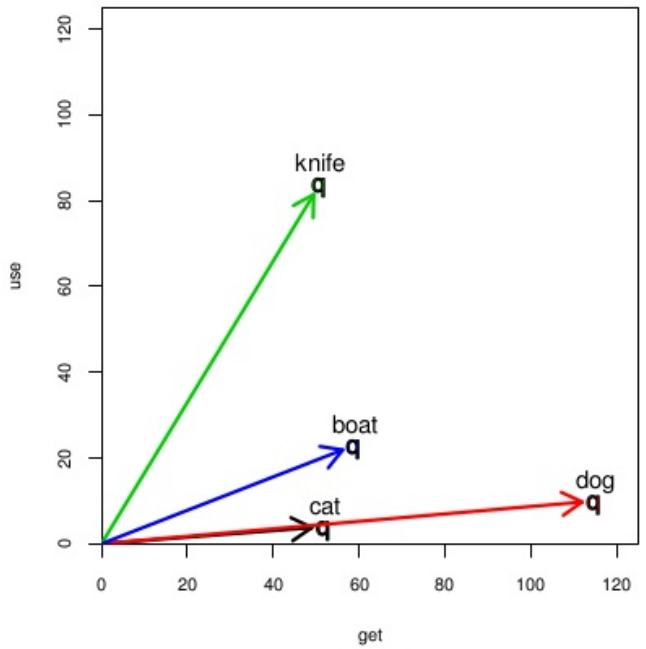


Lexical Semantics



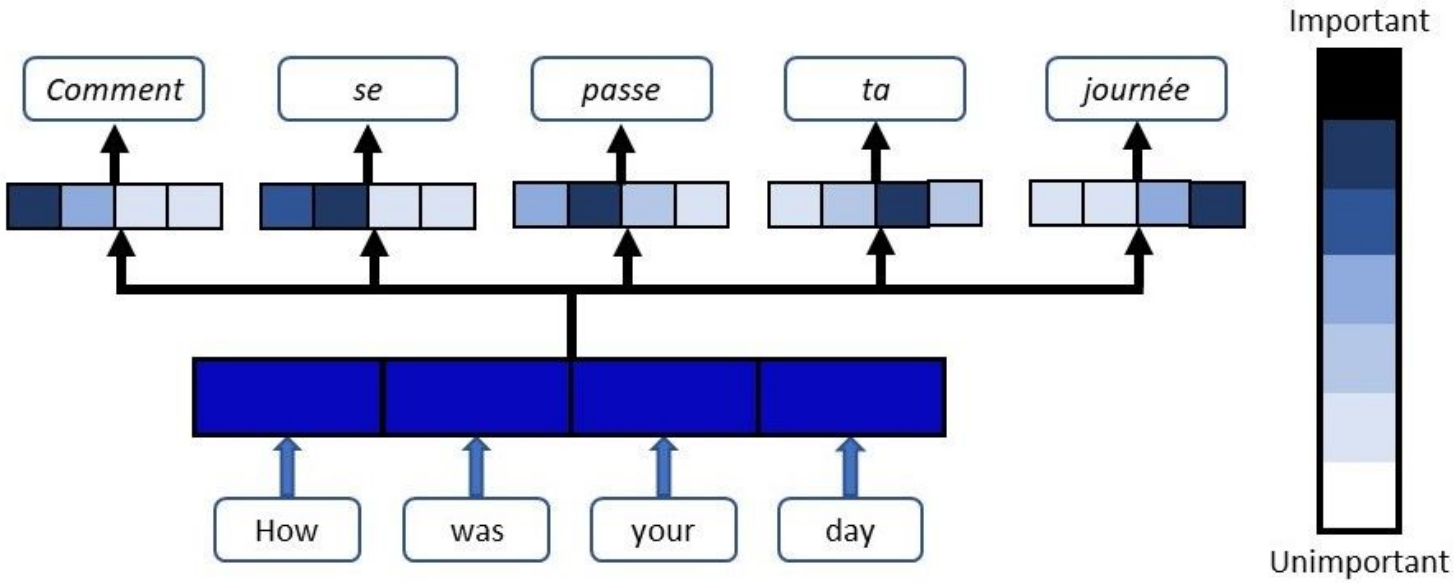
- What is lexical semantics?
- What are word senses?
- What relations exist between words?
- What does the electronic lexical database WordNet contain?
- How can we compute word similarity based on WordNet?

Distributional Semantics



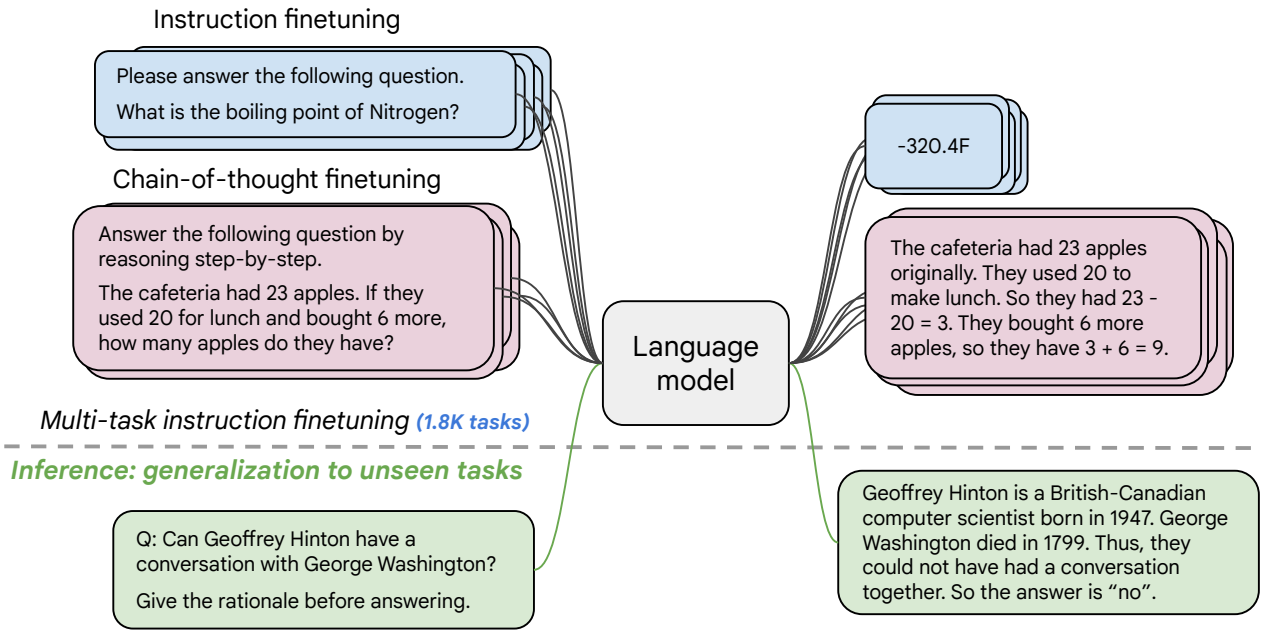
- How can we automatically extract semantic relations between words from texts
- What is distributional semantics?
- What measures and weights are used to compute the similarity between words in texts?
- What is the performance of these automatic methods?

Transformers and the Attention Mechanism



https://blog.floydhub.com/attention-mechanism/

Large Language Models



Information Extraction: Named Entity Recognition

C M Y K Nxxx,2013-06-14,A,001,Bs-BK,E3

"All the News That's Fit to Print"

The New York Times

Late Edition

Today, some sun, a shower or thunderstorm, high 70. **Tonight,** spotty evening thunderstorms, partly cloudy, low 59. **Tomorrow,** sunny, high 79. Weather map, Page B16.

VOL. CLXII . . No. 56,167 © 2013 The New York Times NEW YORK, FRIDAY, JUNE 14, 2013 $2.50



OZAN KOSE/AGENCE FRANCE-PRESSE — GETTY IMAGES

Turkish Protesters Reach Tentative Agreement With Prime Minister
Protesters in Gezi Park in Istanbul on Thursday. Protest leaders held late-night talks with the prime minister in Ankara. Page A4.

U.S. IS SAID TO PLAN TO SEND WEAPONS TO SYRIAN REBELS

Administration Says Assad's Troops Used Chemical Arms in the Civil War

This article is by ***Mark Mazzetti, Michael R. Gordon*** *and* ***Mark Landler****.*

WASHINGTON — The Obama administration, concluding that the troops of President Bashar al-Assad of Syria have used chemical weapons against rebel forces in his country's civil war, has decided to begin supplying the rebels for the first time with small arms and ammunition, according to American officials.

The officials held out the possibility that the assistance, coordinated by the Central Intelligence Agency, could include antitank weapons, but they said that for now supplying the antiaircraft weapons that rebel commanders have said they sorely need is not under consideration.

Supplying weapons to the rebels has been a long-sought goal of advocates of a more aggressive American response to the Syrian civil war. A proposal made last year by David H. Petraeus, then the director of the C.I.A., and backed by the State Department and the Pentagon to supply weapons was rejected by the White House because of President Obama's deep reluctance to be drawn into another war in the

shipments of arms to select groups may be too little, too late.

Some senior State Department officials have been pushing for a more aggressive military response, including airstrikes to hit the primary landing strips in Syria that the Assad government uses to launch the chemical weapons attacks, ferry troops around the country and receive shipments of arms from Iran.

But White House officials remain wary, and on Thursday Benjamin J. Rhodes, one of Mr. Obama's top foreign policy advisers, all but ruled out the imposi-

Continued on Page A6

Justices, 9-0, Bar Patenting Human Genes

In Reprieves, Hint of Immigration Bill's Impact

By KIRK SEMPLE

One found a job as a teacher's assistant at a center for children with autism. Another became a

to the United States as children and are here illegally, allows them to obtain a Social Security number, apply for driver's licenses and work.

As Congress considers an im-

profound. Many talked about being able to walk out their front door for the first time without fear that a wrong turn could lead to deportation proceedings.

"I'm a human here in this

Secret Ruling Put Tech Firms In Data Bind

By CLAIRE CAIN MILLER

SAN FRANCISCO — In a secret court in Washington, Yahoo's top lawyers made their case. The government had sought help in

Information Extraction: Named Entity Recognition

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Entity Linking and Concept Extraction

The majority of patients with Canavan disease are from an Ashkenazi Jewish background . Mutations in ASPA that lead to loss of enzymatic activity have been identified , and E285A and Y231X are the two predominant mutations that account for 97 % of the mutant chromosomes in Ashkenazi Jewish patients .

ACCGTGTACCCCGTGTTTGTGAATG[A/C]GGCCGCATATTACGAAAAGAAAGAA

rs12948217, also known as Y231X or Tyr231Ter, is a SNP in the ASPA gene.

E75.2 Other sphingolipidosis

Disease:

- Fabry(-Anderson)
- Gaucher
- Krabbe
- Niemann-Pick

Farber syndrome

Metachromatic leukodystrophy

Sulfatase deficiency

Relation Extraction

Information extraction pipeline

For years, Microsoft Corporation CEO Bill Gates was against open source. But today he appears to have changed his mind. "We can be open source. We love the concept of shared source," said Bill Veghte, a Microsoft VP. "That's a super-important shift for us in terms of code access."

Richard Stallman, founder of the Free Software Foundation, countered saying...

Name	Title	Organization
Bill Gates	CEO	Microsoft
Bill Veghte	VP	Microsoft
Richard Stallman	Founder	Free Soft..

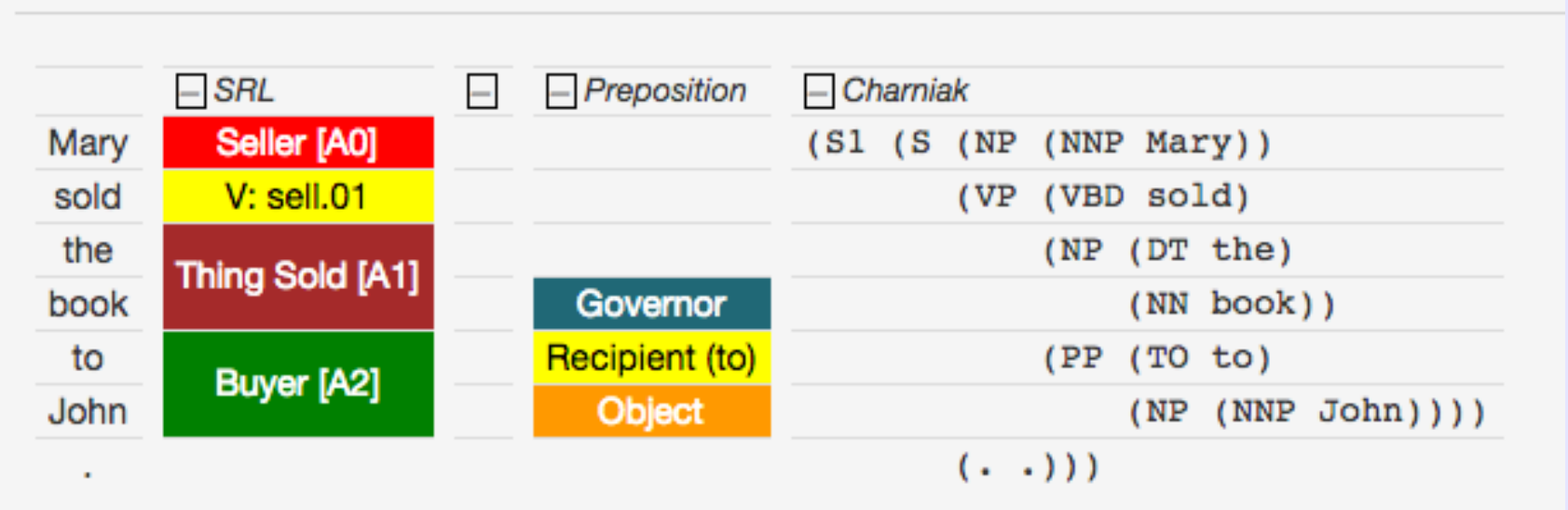
Slide from McCallum

Semantic Role Labeling

Example Sentence

Mary sold the book to John.

SRL output



Coreference Resolution and Entity Network Analysis, Semantic Web, Database Completion

The music was so loud. It couldn't be enjoyed. Our neighbors dislike the music. If they are angry, the cops will show up soon.

```
<div vocab="https://schema.org/" typeof="Person">
  <span property="name">Paul Schuster</span> was born in
  <span property="birthPlace" typeof="Place" href="https://www.wikidata.org/entity/Q1731"
    <span property="name">Dresden</span>.
  </span>
</div>
```

Sentiment Analysis (1)



Sentiment Analysis (2)

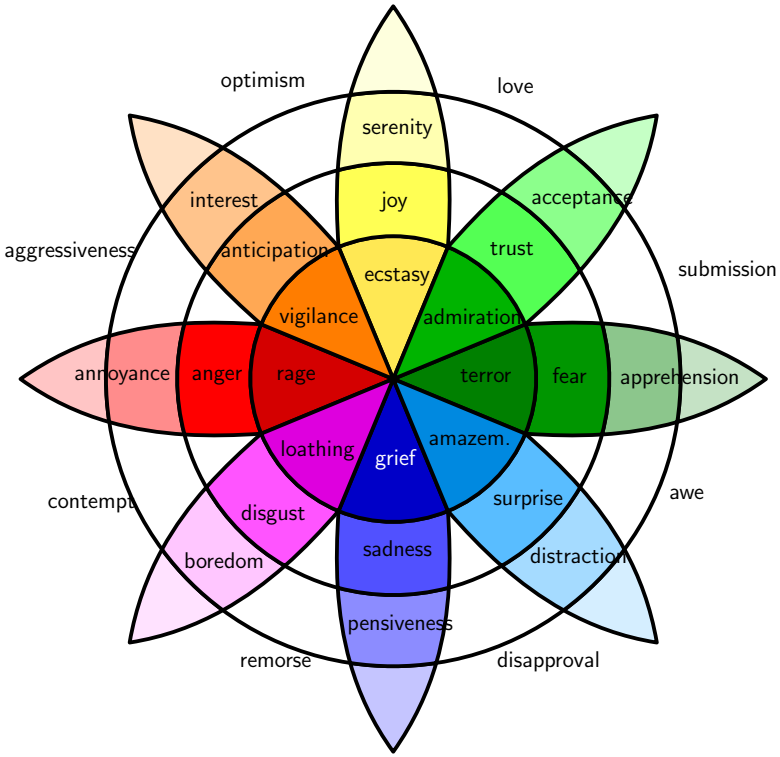
Review for dog costume

So cute! . . . I can tell you it looks great on her and my mom and I got a total kick out of it. You have to cut your own slits for your dog's ears behind the frill, but honestly that's better than having ones in the wrong place, I think. Mine looks a little redder thant what the picture shows, too, but it's just too fabulous, I really don't mind. I cannot wait to walk my dog nearer Halloween. . .

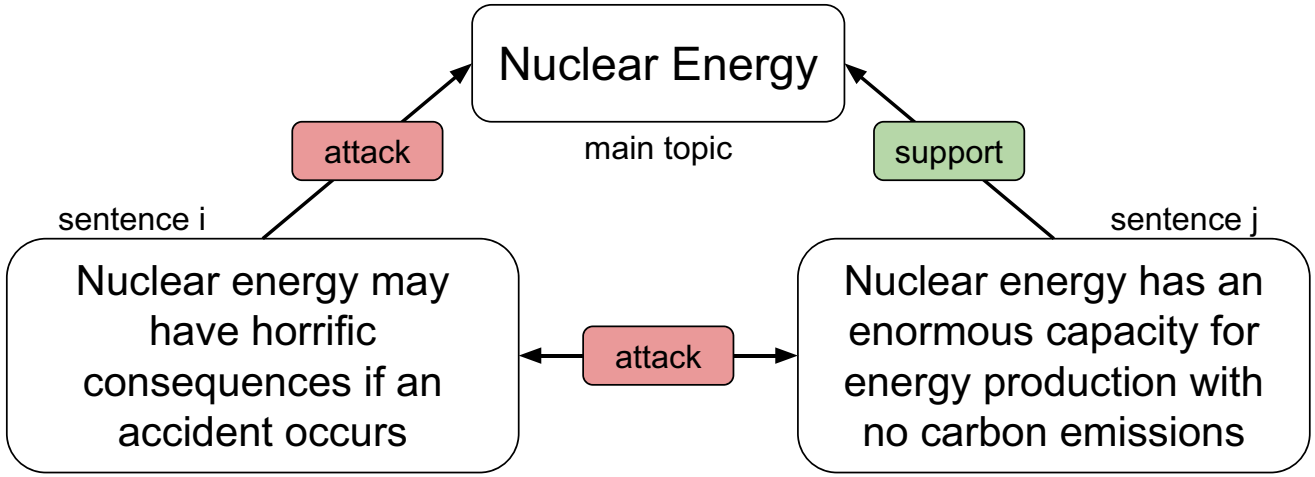
Aspects and Subjective evaluations

- looks – great – got a total kick out of it
- slits – that's better
- frill
- redder – don't mind

Emotion Analysis

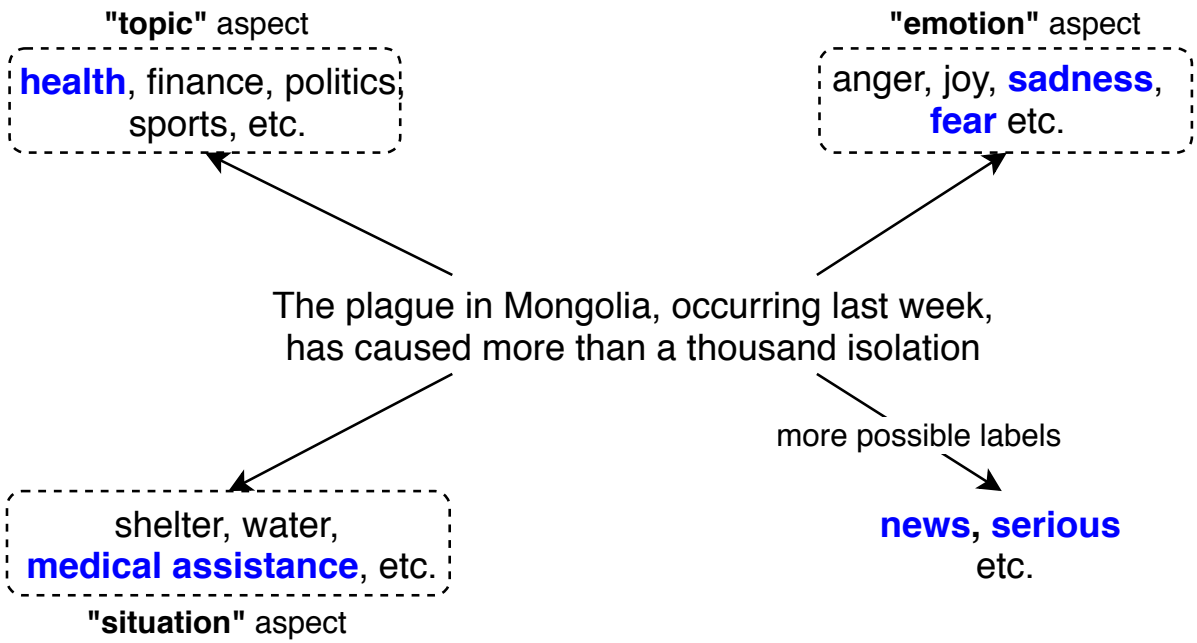


Argument Mining and Fact Checking



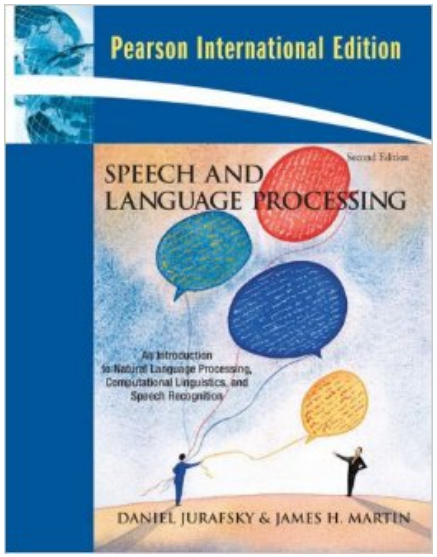
Trautmann et al, 2020: Relational and Fine-Grained Argument Mining

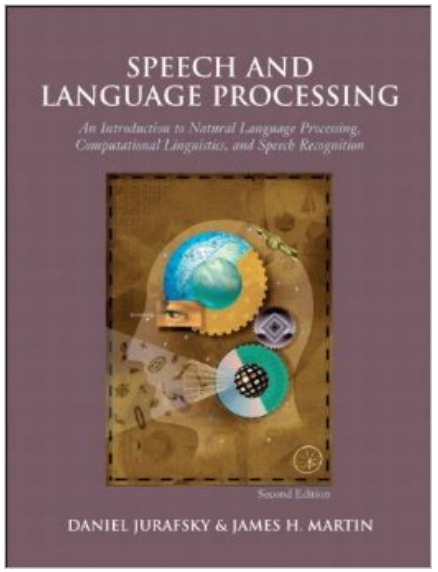
Natural Language Inference and Prompting



Reading Materials

- Jurafsky, Daniel and James H. Martin (2009). *Speech and Language Processing* (2nd Edition). Prentice Hall.





- Other reading material made available during the semester

Time Plan

NLP Planung Wintersemester 2026/2027 (vorläufig, wird sich wahrscheinlich im Verlauf ändern)

Woche	Sitzung	Datum	Wochentag	Person	Thema	Übungsausgabe
1	V1	13.10.2026	Dienstag	K(+W)	Introduction	
	(no meeting)	15.10.2026	Donnerstag			
2	V2	20.10.2026	Dienstag	W	Phonetics	E1-Phonetics
	(no meeting)	22.10.2026	Donnerstag			
3	V3	27.10.2026	Dienstag	W	Document Preprocessing and Tokenization	E2-Prepr-Tok
	E1	29.10.2026	Donnerstag	W	E1-Phonetics	
4	V4	3.11.2026	Dienstag	W	Part of Speech Tagging, Shallow Parsing and Constituency Parsing	E3-ConstPars
	E2	5.11.2026	Donnerstag	W	E2-Prepr-Tok	
5	V5	10.11.2026	Dienstag	W	Dependency Parsing	E4-DepPars
	E3	12.11.2026	Donnerstag	W	E3-ConstPars	
6	V6	17.11.2026	Dienstag	K	Lexical Semantics	E5-LexSem
	E4	19.11.2026	Donnerstag	W	E4-DepPars	
7	V7	24.11.2026	Dienstag	K	Similarity and Distributional Semantics	E6-DistrSem
	E5	26.11.2026	Donnerstag	K	E5-LexSem	
8	V8	1.12.2026	Dienstag	K	Distributional Semantics 2	E7-DistrSem2
	E6	3.12.2026	Donnerstag	K	E6-DistrSem	
9	V9	8.12.2026	Dienstag	K	Compositional and Contextualized Distributional Semantics	E8-Context
	E7	10.12.2026	Donnerstag	K	E7-DistrSem2	
10	V10	15.12.2026	Dienstag	K	Attention und LLM	E9-Attention
	E8	17.12.2026	Donnerstag	K	E8-Context	
11	V11	22.12.2026	Dienstag	K	Information Extraction + NER	E10-NER
	Holidays					
12	Holidays					
	V12	7.1.2027	Donnerstag	K	Relation Extraction + SRL	E11-RE-SRL
13	V13	12.1.2027	Dienstag	K	Entity Linking; Coref	
	E9,E10	14.1.2027	Donnerstag	K	E9-Attention E10-NER	
14	V14	19.1.2027	Dienstag	K	Sentiment, Emotion, Argument	
	V15	21.1.2027	Donnerstag		E11-RE-SRL	
15	Buffer	26.1.2027	Dienstag	W	Ethical Impact of NLP	
	Buffer	28.1.2027	Donnerstag			
	Mock Exam Exercise	2.2.2027	Dienstag	K+W		
	Q&A	4.2.2027	Donnerstag	K+W		

Other topics you would like to be covered?

Send me a mail if something comes up or tell me now.

Take Home

- What is natural language processing about?
- What do we cover in this class?
- What are the formalities?